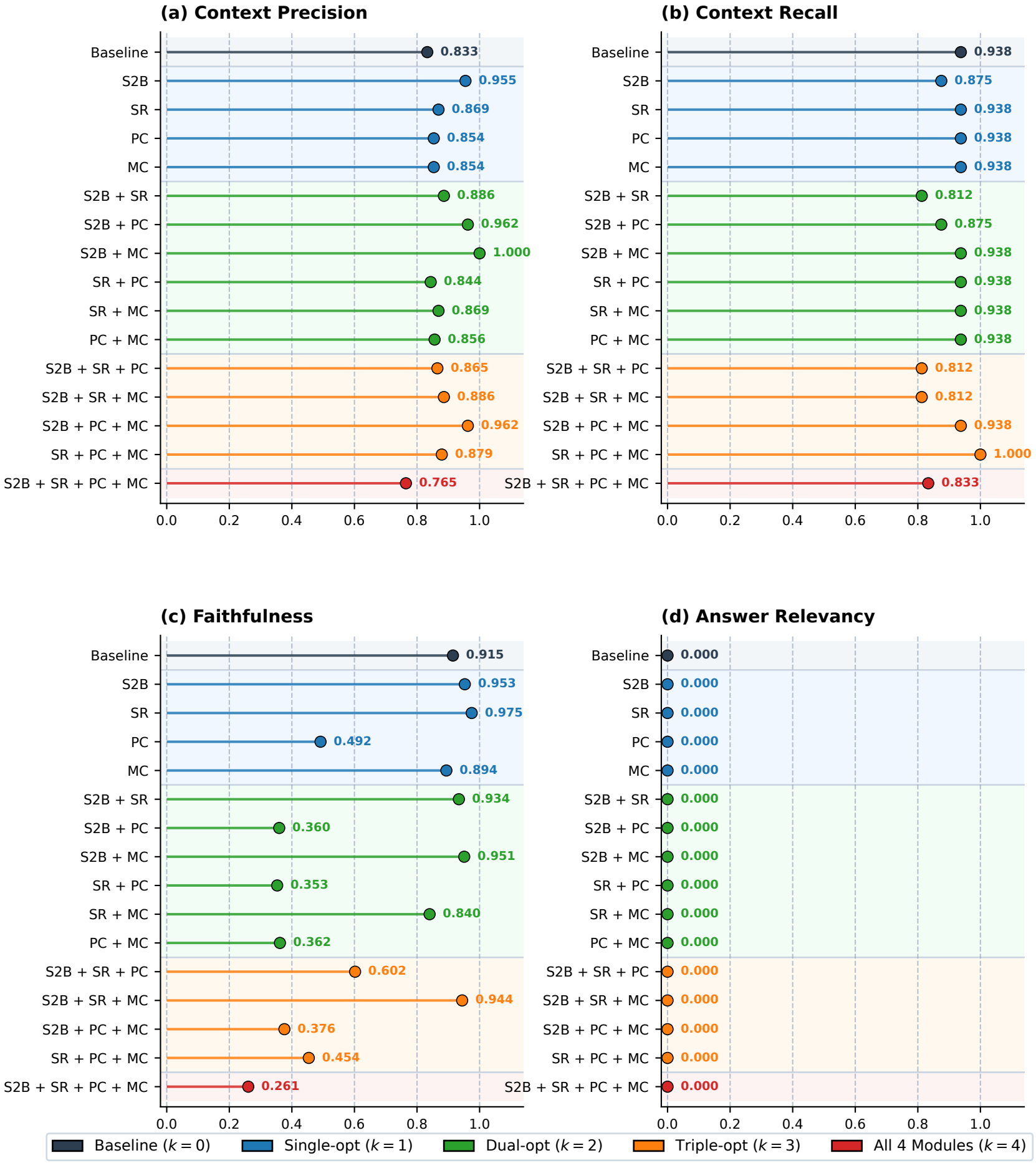
RAGAS Quality Metrics by Pipeline Architecture — Enterprise
(a) Context Precision
Baseline 0.833
S2B 0.955
SR 0.869
PC 0.854
MC 0.854
S2B + SR 0.886
S2B + PC 0.962
S2B + MC 1.000
SR + PC 0.844
SR + MC 0.869
PC + MC 0.856
S2B + SR + PC 0.865
S2B + SR + MC 0.886
S2B + PC + MC 0.962
SR + PC + MC 0.879
S2B + SR + PC + MC 0.765
0.0 0.2 0.4 0.6 0.8 1.0
(b) Context Recall
Baseline 0.938
S2B 0.875
SR 0.938
PC 0.938
MC 0.938
S2B + SR 0.812
S2B + PC 0.875
S2B + MC 0.938
SR + PC 0.938
SR + MC 0.938
PC + MC 0.938
S2B + SR + PC 0.812
S2B + SR + MC 0.812
S2B + PC + MC 0.938
SR + PC + MC 1.000
S2B + SR + PC + MC 0.833
0.0 0.2 0.4 0.6 0.8 1.0
(c) Faithfulness
Baseline 0.915
S2B 0.953
SR 0.975
PC 0.492
MC 0.894
S2B + SR 0.934
S2B + PC 0.360
S2B + MC 0.951
SR + PC 0.353
SR + MC 0.840
PC + MC 0.362
S2B + SR + PC 0.602
S2B + SR + MC 0.944
S2B + PC + MC 0.376
SR + PC + MC 0.454
S2B + SR + PC + MC 0.261
0.0 0.2 0.4 0.6 0.8 1.0
(d) Answer Relevancy
Baseline 0.000
S2B 0.000
SR 0.000
PC 0.000
MC 0.000
S2B + SR 0.000
S2B + PC 0.000
S2B + MC 0.000
SR + PC 0.000
SR + MC 0.000
PC + MC 0.000
S2B + SR + PC 0.000
S2B + SR + MC 0.000
S2B + PC + MC 0.000
SR + PC + MC 0.000
S2B + SR + PC + MC 0.000
0.0 0.2 0.4 0.6 0.8 1.0
Baseline (k = 0)
Single-opt (k = 1)
Dual-opt (k = 2)
Triple-opt (k = 3)
All 4 Modules (k = 4)